



INTELLIGENT ELECTRIC VEHICLE CHARGING NETWORK MANAGEMENT SYSTEM

Agile Scrum Implementation using Jira — Software Engineering Case Study

Course: Software Engineering - CO1 AT-01

Student Name: [VIMALA K]

Industry Context

The rise of EVs is reshaping charging infrastructure needs



Growth of Electric Vehicles

Rapid global EV adoption is driving unprecedented demand for accessible, reliable public charging points.



Challenges in Charging Infrastructure

Operators struggle with fragmented visibility, uneven station utilization, and manual monitoring processes.



Need for an Intelligent Charging Management System

A smart, data-driven platform is needed to connect drivers and operators in real time.

Problem Statement

Growing EV adoption has exposed critical gaps in charging network operations



Long Waiting Times

Drivers queue unpredictably due to lack of visibility into charger occupancy.



Poor Charger Visibility

No real-time view of available, busy, or faulty charging stations.



Inefficient Charger Allocation

Stations are over- or under-utilized without demand-based optimization.



Poor Energy Monitoring

Operators lack consolidated insight into energy consumption and charger health.

Why Agile Scrum?

Chosen over Waterfall for its adaptability to evolving EV-charging requirements

Waterfall vs. Agile

- Rigid, sequential phases
- **Flexible, iterative sprints**
- Requirements fixed upfront
- **Requirements evolve each sprint**
- Feedback only at the end
- **Continuous stakeholder feedback**
- Long delivery cycle
- **Frequent working increments**
- Costly late-stage changes
- **Easy to adapt & maintain**

Key Benefits for This Project



Flexible Requirements



Incremental Development



Continuous Feedback



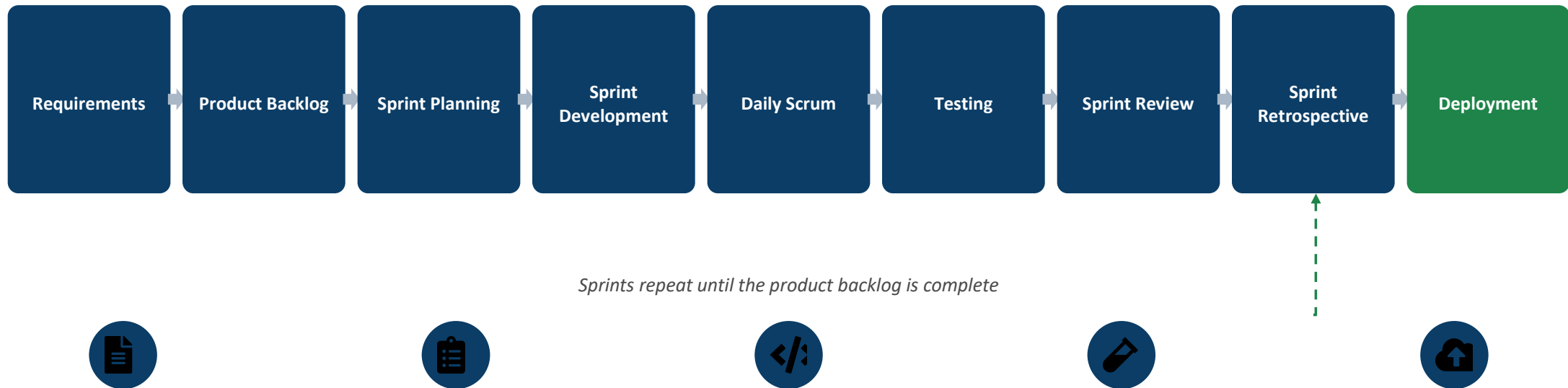
Faster Delivery



Easier Maintenance

Agile Scrum Workflow

End-to-end iterative process followed for the EV charging system



Jira Project Setup

Configuring the Scrum project workspace for the EV charging system



Scrum Project Created

A dedicated Jira Scrum project was set up for the EV charging system.



Product Backlog

All epics and user stories captured and prioritized centrally.



Scrum Board

Visual board tracking issue status across To Do, In Progress, Done.



Sprint Planning

Backlog items pulled into time-boxed sprints (2 weeks each).



Issue Types

Structured as Epics → Stories → Tasks for full traceability.

The screenshot displays the Jira Scrum project workspace for the "Intelligent EV Charging Network Management System". The interface includes a top navigation bar with a search bar, a "+ Create" button, a "Premium trial 17d" badge, and a "Ask Rovo" button. Below the navigation bar, the "Spaces" section shows the project name and a list of views: Summary, List, Board (selected), Backlog, Calendar, Timeline, Docs, and Forms. The "Board" view is active, showing a Kanban board with four columns: "TO DO 2", "IN PROGRESS 1", "IN REVIEW 1", and "DONE 1". Each column contains one or more issue cards. The "TO DO" column has two cards: "Connect Database" (REPORT AND DASHBOARD, due Jul 28, 2026, assigned to IECNMS-17) and "Validate candidates" (NOTIFICATION SYSTEM, due Jul 28, 2026, assigned to IECNMS-18). The "IN PROGRESS" column has one card: "Create Login Page" (CHARGING STATION MANAGEMENT, due Jul 28, 2026, assigned to IECNMS-16). The "IN REVIEW" column has one card: "Registration User Database" (ENERGY MONITORING, due Jul 29, 2026, assigned to IECNMS-9). The "DONE" column has one card: "User Login" (USER MANAGEMENT, due Jul 28, 2026, assigned to IECNMS-15). Each card also shows the assignee's initials and a status icon. A "+ Create" button is located at the bottom left of the board.

Product Backlog

Epics prioritized in Jira for the EV charging management system

Spaces

Intelligent EV Charging Network Management System

...

Summary

List

Board

Backlog

Calendar

Timeline

Docs

Forms

+

VK

AV

MS

Filter

☐ IECNMS Sprint 1 27 Jul – 10 Aug (5 work items)

000 Complete sprint

Develop Login and Registration Module

☒ IECNMS-9 Registration User Database	ENERGY MONITIING	IN REVIEW	Jul 29	-	=	MS
☐ IECNMS-15 User Login	USER MANAGENMENT	DONE	Jul 28	-	>	VK
☒ IECNMS-16 Create Login Page	CHARGING STATION ...	IN PROGRESS	Jul 28	-	=	AV
☒ IECNMS-17 Connect Database	REPORT AND DASHBO...	TO DO	Jul 28	-	<	MS
☒ IECNMS-18 Validate candiates	NOTIFICATION SYST...	TO DO	Jul 28	-	=	VK

+ Create

Jira

Q Search

+

Create

Premium trial 17d

Ask Rovo

?

For you

Recent

Starred

Apps

Explore more apps

Plans

Once you open or create plans, they'll show up here.

View all plans

Spaces

Recent

Intelligent EV Charging Network...

Kanban work flow

My Software Team

(Example) Billing System Dev

More spaces

Recommended

Prioritize ideas

Spaces

Intelligent EV Charging Network Management System

SummaryListBacklogBoardCalendarTimelineDocsForms+

Q Search timeline

VKMSAV

Epic

Status category

Work

ne

July

August

Sprints

IECNMS Sprint 1

IECNMS-2 user management

IECNMS-3 Charging Station Management

IECNMS-4 Booking Management

IECNMS-5 Waiting time prediction

IECNMS-6 Energy Monitoring

IECNMS-7 Report and dashboard

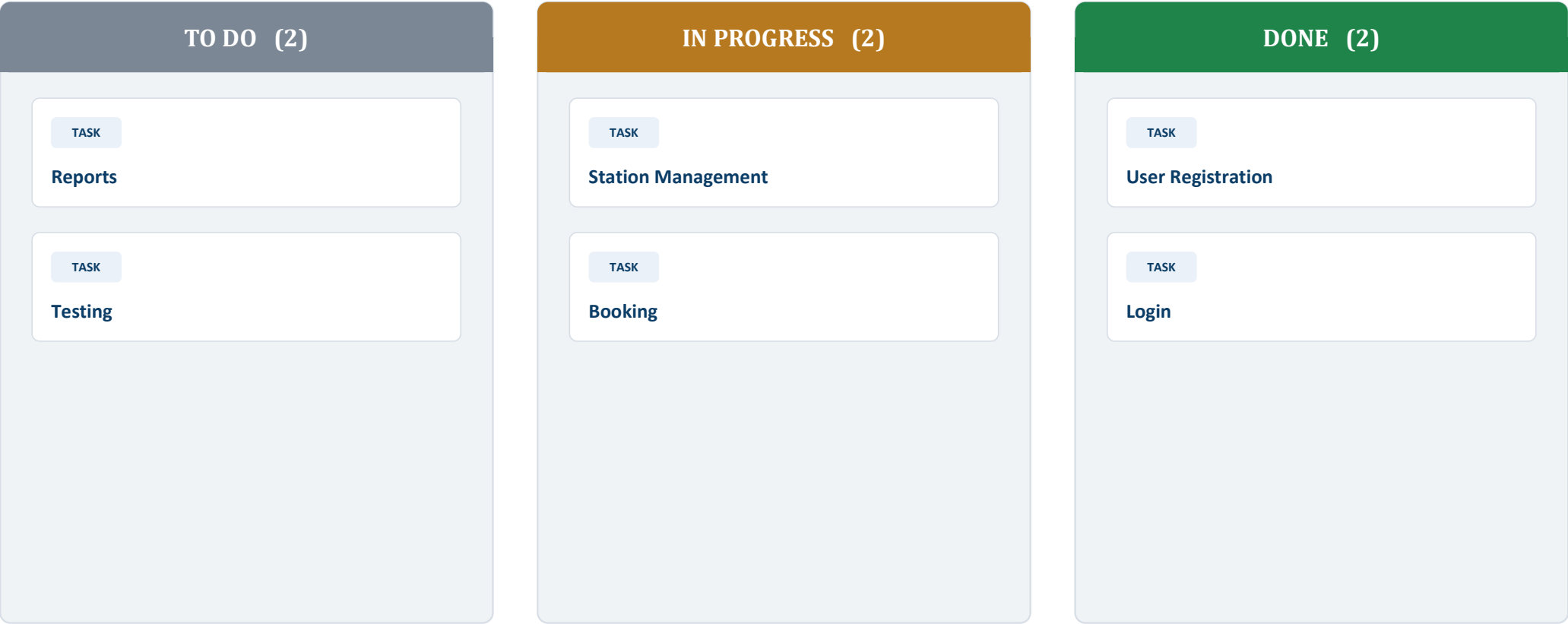
IECNMS-8 Notification System

+ Create Epic

TodayWeeksMonthsQuarters

Jira Scrum Board

Live sprint task tracking across three workflow columns



Intelligent EV Charging Network Management System

Summary List Board Backlog Calendar Timeline Docs Forms +

Filter

**1 completed**

in the last 7 days

**17 updated**

in the last 7 days

**17 created**

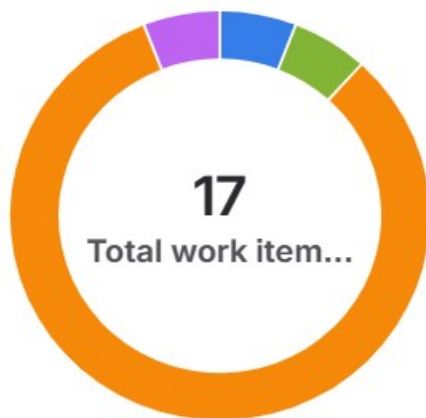
in the last 7 days

**15 due soon**

in the next 7 days

Status overview

Get a snapshot of the status of your work items. [View all work items](#)



In Progress: 1

Done: 1

To Do: 14

In Review: 1

Recent activity

Stay up to date with what's happening across the space.

Today



Vimala Kv changed the Priority from

Medium to High on

25 minutes ago



IECNMS-17: Connect Database

TO DO



Vimala Kv changed the Priority from

Medium to Low on

25 minutes ago



IECNMS-15: User Login

DONE



Vimala Kv changed the Priority from

Medium to Highest on

25 minutes ago



IECNMS-12: Waiting Time Prediction Notifications

TO DO

Expected Outcomes

Measurable impact of the Agile-delivered EV charging platform



Reduced Waiting Time



Better Charger Utilization



Improved Customer Satisfaction



Real-Time Monitoring



Energy Optimization



Efficient Station Management



Conclusion

Adopting Agile Scrum with Jira enabled the team to break the EV Charging Network Management System into manageable epics and sprints, incorporating continuous feedback and delivering working increments every cycle. This iterative approach reduced risk, kept requirements aligned with real driver and operator needs, and ensured a scalable, well-tested platform reached deployment efficiently.

- ✓ Transparent sprint-based progress tracking in Jira
- ✓ Faster response to changing requirements
- ✓ Higher quality through continuous testing & review
- ✓ Successful, on-time incremental delivery